



Using Unsupervised Methods to Classify X-ray Fluorescence Data

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Refresher

- A pXRF device is a device that fires x-rays at materials and can determine what the chemical composition of that material is
- Larry's capstone project currently uses supervised methods to assign labels to samples
- This project aims to use unsupervised methods to cluster samples together and to label them
- We are using autoencoders and HBDSCAN to reduce the dimensionality of the data and cluster like samples together



Technical Implementation And Correctness

In order to implement a pipeline that should accomplish the sample clustering, we followed these steps:

- Preprocess the data
- Scale the element concentrations
- Use autoencoders to turn the 18/34 dimensional data into 8 dimensional data
- Use HBDSCAN to cluster like samples together
- For Dr. Sharp's dataset, we can directly test the accuracy



Experimental Design and Evaluation

To make sure our experiment was sound, we:

- Had an 80/20 split and balanced the set
- Used autoencoders + HBDSCAN to cluster the samples
- We prevented overfitting using early stopping with a patience of 10 and restoring best weights

To evaluate the model, we:

- Used silhouette, basic accuracy and F1-score
- Used majority vote for mapping clusters to classifications
- Confusion matrices for visualizing where the model missed



Contribution and Insight

- The previous implementation used Gradient Boosted Decision Tree, which relies on labeled data to create a decision tree where each sample would traverse it until it fell into a category (soil, slag, metal pottery).
- Our new method is to use autoencoders and HBDSCAN to try and cluster the samples with high accuracy. Since most pXRF data isn't labeled, exclusively using supervised methods for labeling samples isn't a reliable strategy. Being able to group samples together before labeling would save researchers a ton of time.



Reproducibility and Documentation

- Everything is ran through either “Autoencoder HDBscan Labeled.py” or “Autoencoder HDBscan Unlabeled.py”
 - Arguments include the csv input, an output directory as standard arguments
 - Additionally, latent space dimensionality, # of epochs, batch size, test size, specific feature list, or target column can be specified
- The entire pipeline is handled through this program
- Results are returned in either the “results” directory, or a specified directory
 - Figures are generated at the end of the pipeline and are stored in the specified results directory
- Our website is located [here](#)



Deliverable Quality

- We opted to create a website, as it gives us a clean way to show our results and our work process
- Our website acts as an overview of our project, including:
 - An introduction to X-Ray Fluorescence and how unsupervised models can help
 - An explanation of our workflow
 - A list of our datasets, what they are, and where they come from
 - Our results from scanning the datasets
- All this creates a clear and cohesive picture of our project and what it accomplished



GitHub Repository

Repository: <https://github.com/LunaticSage218/CS472FinalProject.git>

- Complete with a README.md file that includes how to setup the virtual environment, so there's no conflicting library versions
- Includes a requirements.txt for importing packages
- MIT License



Website

<https://mjh692.z5.web.core.windows.net/CS472FinalProject>



Thank you!

Any Questions?